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On the isolated impact of copulas on risk measurement: A simulation study

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ABSTRACT

This paper quantifies the impact of fundamental copula approaches on applied risk measurement with particular focus on Value-at-Risk (VaR) forecasts.The application of a simulation study reveals the impact of misspecified dependence modeling on VaR forecasts. In particular, accounting for several degrees of joint extreme movements and time varying dependence of the simulated return series, it is the t copula that describes a robust approach to achieve adequate VaR forecasts.

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1. Introduction

Triggered by the financial crisis, modeling dependence between financial time series represents a growing field of research. Especially in turmoil market times, dependence measurement is impacted by the properties of the underlying return series and becomes crucial for applied risk management (see Ane and Kharoubi, 2003, Angelidis et al., 2004 and Fantazzini, 2009).

Due to the fact, that the linear correlation coefficient lacks in capturing nonlinear transformations of marginal return distributions (Cherubini et al., 2004), copulas have become a promising alternative in the field of finance.

In this vein, Embrechts et al. (2002) introduced the application of basic copula approaches to financial return series and Patton (2004) expands the copula framework with respect to the time varying nature of financial dependence schemes. Halbleib and Pohlmeier (2012); Chen and Tu (2013) and Ane and Kharoubi (2003) assess copulas in the field of forecasting and Aloui et al. (2011); Berger and Missong (2013) and Chen and Tu (2013) underline the superiority of copulas in comparison to linear correlation coefficients for applied risk measurement. Moreover, according to several quality criteria, Patton (2004); Aloui et al. (2011) and Boubaker and Sghaier (2013) favor the application of the Gumbel copula to financial data. However, the evaluation of adequate dependence modeling depends on the applied quality tests and the underlying sample characteristics do heavily impact the performance of the applied copulas (see also Junker and May, 2005; Huang et al., 2009; Hsu et al., 2011; Ozun and Cifter, 2007 and Kole et al., 2007 amongst others). Thus, the adequate choice of copula describes a staggering task, especially in periods of financial turmoil (Grundke and Polle, 2012).

With particular focus on Value-at-Risk (VaR), He and Gong (2009) use copulas to couple credit and market risk and constitute an adequate quality of VaR forecasts based on copulas. As a result, no remarkable differences between standard copula approaches (i.e. Gauss, t, Clayton, Frank and Gumbel copulas) is stated. Berger (2013) applies copulas to stocks, indices and foreign exchange rates and Chen and Tu (2013) apply copulas to index futures and spot returns that lead to adequate VaR forecasts. Although, both surveys underline the superiority of copulas towards classical parametric approaches, the impact of competing copula approaches seems to be of marginal importance for the quality of VaR forecasts. According to Ane and Kharoubi (2003), copulas explain up to 18% of the absolute portfolio variance and, as presented by Fantazzini (2009), marginal return distributions heavily impact dependence measurement and describe the main impact on portfolio volatility.

In this paper, we add to the discussion on adequate dependence modeling for financial portfolios and quantify the impact of misspecified dependence measurement on VaR forecasts. Due to the impact of marginal return distributions, it is not straightforward to isolate the impact of misspecified dependence modeling from the impact of misspecified marginal return distributions for real data. Thus, we set up a simulation study that enables us to assess the isolated impact of dependence modeling on the quality of VaR forecasts.

We apply basic copula approaches in an applied VaR setup and quantify the impact of each approach with focus on applied risk measurement. Based on realistic samples that are characterized by

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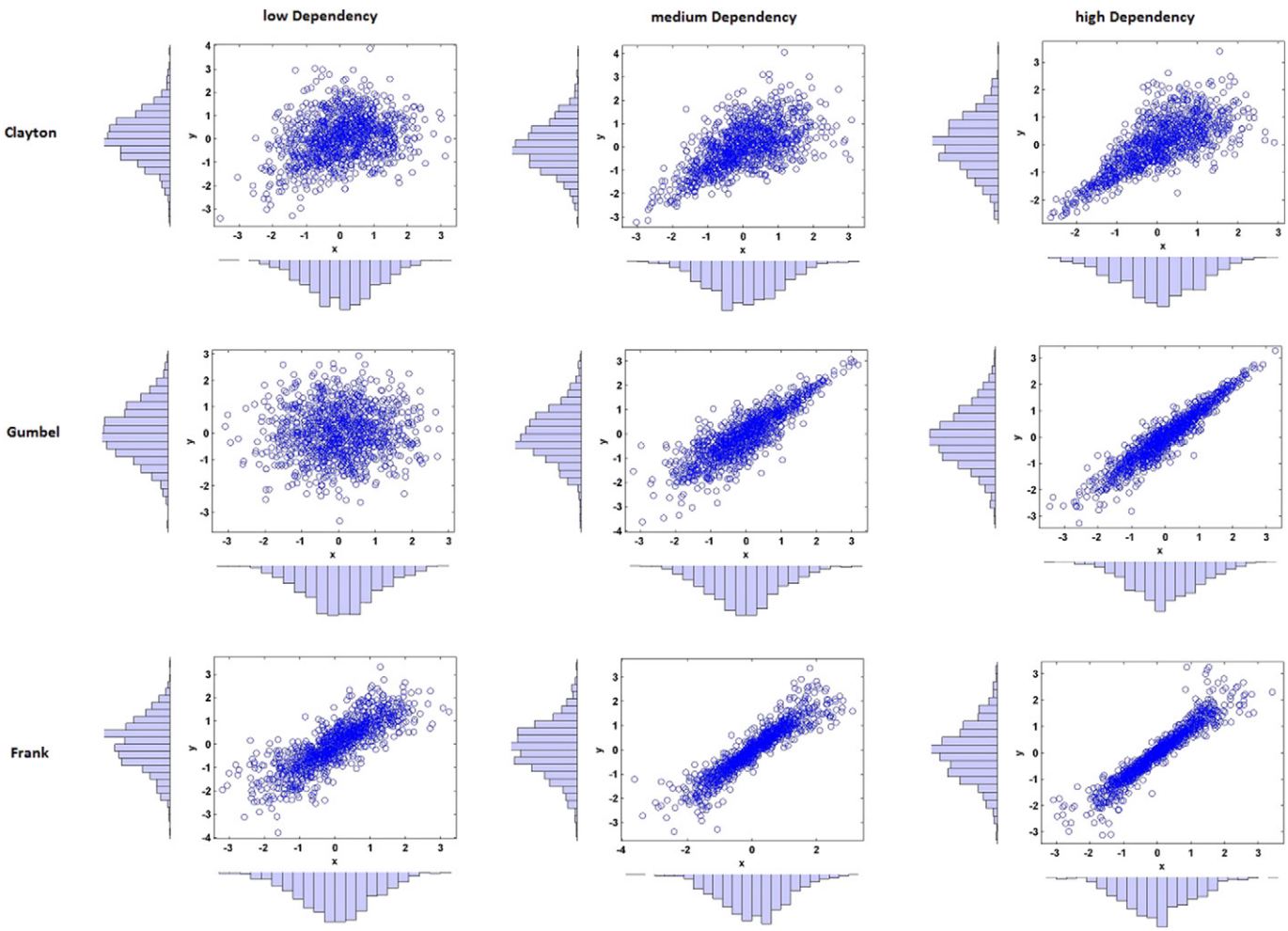

Fig. 1. 1001 normally distributed return observations generated by Clayton, Gumbel and Frank copula.

different dependence schemes (tail dependence and time varying dependence) we assess the isolated impact of competing copula approaches on the backtesting performance of daily VaR forecasts.

The remainder of this paper is structured as follows: Section 2 describes the methodology and Section 3 presents the simulation setup and the respective results. Section 4 concludes.

2. Methodology

2.1. Copulas

The copula approach goes back to Sklar (1959), who introduces the underlying assumptions of copulas and proves the existence of joint probability functions. Based on Sklar (1959), copula dependence is modeled apart from the choice of univariate marginal return distribution and the bivariate joint probability function is defined as:

$$F(x_1, x_2) = C(F_1(x_1), F_2(x_2)). \tag{1}$$

In our simulation analysis, x_i denotes the simulated return series of asset $i = 1, 2$ and F_i the respective marginal distribution. Further, if the univariate marginal distributions are known, the copula dependence (C) can be estimated.

Before we fit the copula parameters of the respective copula approaches to our data, we transform the simulated return series to the copula scale by applying the cumulative distribution function (cdf) of the univariate marginal return distribution. In the remainder, we will provide a brief introduction to the applied copula approaches that will be exclusively discussed in the context of normally distributed asset returns[1]. For a thorough introduction to copula modeling we refer to Cherubini et al. (2004) and Manner and Reznikova (2012).

Both Gaussian and t copula belong to the family of elliptical copulas. That is, both copulas are derived from the elliptical multivariate normal and t distribution and the setup of the Gaussian copula is given by:

$$C^{Ga}(u_1, u_2) = \Phi_\rho(\Phi^{-1}(u_1), \Phi^{-1}(u_2)), \tag{2}$$

$$= \int_{-\infty}^{\Phi^{-1}(u_1)} \int_{-\infty}^{\Phi^{-1}(u_2)} \frac{1}{2(\pi)^{\frac{n}{2}} |\rho|^{\frac{1}{2}}} \exp\left(-\frac{1}{2} z^T \rho^{-1} z\right) dz_1 dz_2, \tag{3}$$

with Φ_ρ representing the multivariate normal distribution with correlation matrix ρ and Φ^{-1} as the inverse of univariate normal

[1] It is to note, that the choice of elliptical marginal return distribution does not impact the presented results.

distribution. As the simulated return series are described by unviariate normal distributions, the setup of the Gaussian copula is similar to the setup of the multivariate Gaussian distribution. It is to note, that in our simulation setup, due to normally distributed returns, both Gauss copula and linear correlation lead to identical results. Therefore we just present the results for the Gauss copula.

Analogous to the setup of the Gaussian copula, the t copula draws on the multivariate t distribution and is given by:

$$C^t(u_1, u_n) = t_{\rho,v}(t_v^{-1}(u_1), t_v^{-1}(u_2)), \tag{4}$$

$$= \int_{-\infty}^{t^{-1}(u_1)} \int_{-\infty}^{t^{-1}(u_2)} \frac{\Gamma\left(\frac{v+n}{2}\right)}{\Gamma\left(\frac{v}{2}\right)(v\pi)^{\frac{n}{2}}|\rho|^{\frac{1}{2}}} \left(1 + \frac{1}{v} z^T \rho^{-1} z\right)^{-\frac{v+n}{2}} dz_1 dz_2. \tag{5}$$

In this setup $t_{\rho,v}$ stands for the multivariate t distribution with correlation matrix ρ and v degrees of freedom (d.o.f.). t_v^{-1} stands for the inverse of the univariate t distribution and v determines the degree of symmetric tail dependence. If the individual return series follow univariate t distributions that are characterized by identical degrees of freedom, the t copula equals the multivariate t distribution. However, in contradiction to the multivariate t distribution, the setup of the t copula allows for individual degrees of freedom for each asset, as well as for individual return distributions.

Additionally, to assess different patterns of tail dependence, we apply Archimedean copulas. Generally, Archimedean copulas allow for different tail dependence schemes and we discuss three basic approaches that describe positive, negative and symmetric tail dependence (see Cherubini et al., 2004).

The Clayton copula presents a copula approach that explicitly focuses on the joint probability of negative returns and its setup is given as follows:

$$C^{Clayton}(u_1, u_2) = \left(\max\{u_1^\theta + u_2^\theta - 1, 0\}\right)^{\frac{1}{\theta}}, \tag{6}$$

with $\theta \in [-1, \infty)\backslash\{0\}$. That is, for $\theta \to \infty$ the Clayton copula describes strong negative tail dependence, and for $\theta \to 0$ independence.

In contradiction to the Clayton copula, the Gumbel copula describes positive tail dependence and is defined as:

$$C^{Gumbel}(u_1, u_2) = \exp\left(-[(-ln\, u_1)^\theta + (-ln\, u_2)^\theta]^{\frac{1}{\theta}}\right), \tag{7}$$

with $\theta \in [1, \infty)$. Again, $\theta \to \infty$ leads to strong tail dependence, and $\theta \to 0$ presents independence.

Additionally, we introduce the Frank copula that captures symmetric tail dependence. As defined by Nelson (1999) the methodological setup is given by:

$$C^{Frank}(u_1, u_2) = -\frac{1}{\theta} ln\left(1 + \frac{(e^{-\theta u_1} - 1)(e^{-\theta u_2} - 1)}{e^{-\theta} - 1}\right), \tag{8}$$

for $\theta \in \mathbb{R}\backslash\{0\}$. It is to note that all Archimedean copulas are described by one parameter θ, which makes dependence modeling for more than two assets difficult.

Further, to separate dependence modeling from the choice of univariate marginal return distribution, all parameters will be estimated in a two-step maximum likelihood method as introduced by Joe (1996). The likelihood estimation gets divided into one term incorporating all parameters concerning univariate margins and into one term involving the parameters of the chosen copula. Thus, this setup allows for an assessment of the different copula parameters (second step), based on a unified univariate setup (first step).

2.1.1. VaR

To quantify the impact of competing copula approaches on portfolio risk, we translate portfolio risk into a VaR universe. Then, based on daily VaR forecasts, we are able to evaluate the properties of different copulas within a realistic risk measurement framework.

Generally, VaR is defined as the maximum loss limit of the underlying portfolio returns that will not be exceeded with a given probability (α). Typically, probabilities between $1 - \alpha = 95\%$ and $1 - \alpha = 99\%$ are chosen for a holding period of 1 day:

$$\alpha = Pr[r_{t+1} < V\hat{a}R_{t+1}(\alpha)] = F(V\hat{a}R_{t+1}(\alpha)), \tag{9}$$

with r_{t+1} as the return at time $t + 1$ and $F(x)$ as the cdf of r_{t+1}. Thus, for daily return series, the expected VaR at time $t + 1$ ($V\hat{a}R_{t+1}$) is defined as the quantile at level α of the distribution of daily return series.

Therefore, based on the fitted copulas we simulate 10.000 returns and the portfolio VaR at time $t + 1$ is described by the empirical quantile of the vector of simulated portfolio returns based on the information available at time t (see Fig. 2). Moreover, to avoid any offsetting effect of misspecified dependence, we discuss bivariate and equally weighted portfolios.

2.1.2. Backtesting

In order to evaluate the VaR performance of the simulated portfolios, let a VaR breach be defined by a realized loss that exceeds the respective VaR forecast:

$$I = \begin{cases} 1 & if \quad r_{t+1} < V\hat{a}R_{t+1} \\ 0 & if \quad r_{t+1} \geq V\hat{a}R_{t+1}. \end{cases} \tag{10}$$

I presents the relevant indicator variable that indicates a breach by $I = 1$ and a return within the VaR limit by $I = 0$. Consequently, the applied models can be compared by their empirical rate of VaR breaches in an out-of-sample analysis. For instance, for 99% VaR forecasts, we expect to observe 1% of VaR breaches. Hence, if we analyse daily VaR forecasts for one year (250 trading days), we expect to observe either 2 or 3 breaches.

With particular focus on the empirical VaR performance, the relevant statistical quality criterion for our simulation analysis is given by Kupiec (1995). Namely, we apply the backtesting criteria of unconditional coverage (UC) and we test the null hypothesis, if

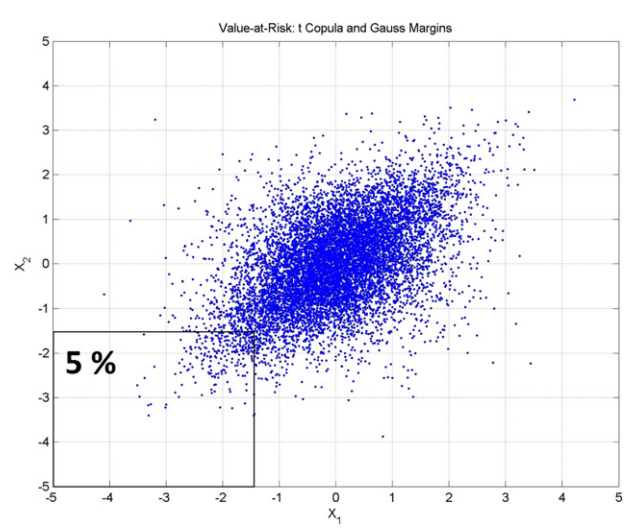

Fig. 2. t copula with Gaussian margins: Empirical 95% Value-at-Risk.

Table 1
Archimedean copulas: Parameters of the simulated scenarios.

Copula	Low dependence	Medium dependence	High dependence
Clayton	$\theta = 0.5$	$\theta = 1.5$	$\theta = 2.5$
Gumbel	$\theta = 1$	$\theta = 3$	$\theta = 5$
Frank	$\theta = 10$	$\theta = 20$	$\theta = 30$

the realized failure rate is in line with the expected failure rate. The relevant test statistic (LR_{UC}) is given by:

$$LR_{UC} = -2ln[(1-p)^{T-N}p^N] + 2ln[(1-(N/T))^{T-N}(N/T)^N]. \quad (11)$$

At this, p stands for the percent left tail level, T for the total days and N for the number of breaches. The null is given by $H_0 : p = N/T$ whereas the test statistic is $\chi^2(1)$ distributed.

As this backtesting procedure incorporates the official Basel backtesting rules it represents a widely accepted benchmark (see Aloui et al., 2011Chen and Tu, 2013; Berger and Missong, 2013).

3. Simulation exercise

First, we investigate tail dependence and simulate samples with each of the analyzed copula to examine the compatibility of the applied copula approaches. As each copula describes joint movements in a different manner, we are able to study the impact of the underlying assumptions on each copula and its corresponding VaR forecasts. Second, we take account of realistic dependence schemes that can not be captured by any of the basic copula concepts and simulate time varying dependence schemes.

3.1. Simulation design

In the following, the simulation setup of the analysis of tail dependence will be introduced stepwise:

In order to assess three patterns of tail dependence, we generate samples via Clayton, Gumbel and Frank copula. Further, for the three different copulas, we investigate scenarios that are characterized by weak, medium and strong dependence. The respective parameterizations are presented in Table 1.

As daily VaR forecasts are mainly modeled by a data window not larger than 1000 observations (see Angelidis et al., 2004, Aloui et al., 2011 and Berger, 2013), we intend to set up a realistic framework and set the sample size to 1001 observations. Moreover, we choose a bivariate portfolio setup to avoid offsetting effects on a portfolio level. Furthermore, to discard any bias from misspecified margins, we assess normally distributed returns.

Thus, for each copula and each parameterization, we simulate two normally distributed return series (1001 observations) that are characterized by the tail dependence of the chosen copula approach. Table 1 exemplifies the nine scenarios.

Next, in order to forecast VaR based on the simulated data, we take the first 1000 returns as our sample and fit the competing copulas to forecast 95% and 99% VaR. Table 2 illustrates 95% VaR forecast, based on a t copula. As the marginal distribution of the returns is assumed to be normal, the difference of the VaR forecasts exclusively stems from dependence modeling.

Further, we evaluate the VaR forecast by comparing each forecast (based on Gauss, t, Gumbel, Clayton or Frank copula) with the 1001st observation of the simulated sample. Consequently, assess five competing 95% and 99% VaR forecasts for each simulation.

Then, to avoid random drawbacks, we repeat the simulation and the corresponding forecasting exercise 10.000 times for each copula and each parameterization. In doing so, we are able to assess the quality of 10.000 VaR forecasts based on each copula for every dependence scheme Graph 3 illustrates the empirical failure rate of 1000 VaR forecasts. Based on the given setup, we are able to judge the quality of the investigated approach by comparing the expected failure rate with the empirical failure rate.

Consequently, for each parameterization (as given in Table 1), we simulate 10.000 samples and evaluate five copulas for each sample. In total, we discuss 450.000 VaR forecasts for 90.000 samples that are

Table 2
Tail dependence: Empirical performance of 95% and 99% VaR forecasts.

Copula	Scenario	G Cop	t Cop	Clayton	Gumbel	Frank	G Cop	t Cop	Clayton	Gumbel	Frank
Clayton	$\theta = 0.5$	6.12%**	6.18%*	5.35%	7.00%**	6.47%**	1.75%**	1.75%**	1.17%	2.32%**	2.18%**
Clayton	$\theta = 1.5$	6.22%**	6.18%**	5.08%	7.21%**	6.48%**	1.54%*	1.41%*	1.05%	2.19%**	2.16%**
Clayton	$\theta = 2.5$	5.97%*	5.91%	5.04%	6.94%**	6.16%**	1.47%*	1.30%	0.97%	2.02%**	2.05%**
Gumbel	$\theta = 1$	4.99%	4.94%	4.88%	4.96%	5.00%	1.08%	0.99%	0.91%	1.01%	1.00%
Gumbel	$\theta = 2$	4.57%	4.63%	4.52%	4.82%	4.81%	0.79%	0.70%	0.65%	0.96%	1.20%
Gumbel	$\theta = 3$	4.57%	4.52%	4.41%	4.68%	4.49%	0.71%	0.63%	0.64%	0.78%	1.10%
Frank	$\theta = 10$	5.08%	4.98%	4.79%	5.66%	5.05%	0.52%**	0.56%**	0.45%**	0.86%	0.95%
Frank	$\theta = 20$	4.92%	5.04%	4.75%	5.28%	4.65%	0.52%**	0.50%**	0.43%**	0.67%	0.80%
Frank	$\theta = 30$	5.20%	5.20%	5.14%	5.39%	5.05%	0.71%	0.71%	0.65%	0.90%	0.99%

Rejection of LR_{UC} test statistics: **/*=99%/95% significance.

Table 3
Time varying dependence: Empirical performance of 95% VaR and 99% VaR forecasts.

Scenario	Dependence	G Cop	t Cop	Clayton	Gumbel	Frank	G Cop	t Cop	Clayton	Gumbel	
$\rho = 0.8$	Constant	4.78%	4.80%	4.48%	5.57%	5.12%	0.98%	0.97%	0.80%	1.39%	1.62%**
$\rho = 0.8$	Persistent	5.00%	5.00%	4.70%	5.67%	5.29%	0.91%	0.91%	0.80%	1.31%	1.66%**
$\rho = 0.8$	Volatile	5.20%	5.10%	4.73%	5.62%	5.34%	1.12%	1.03%	0.89%	1.50%	1.70%**
$\rho = 0.3$	Constant	4.79%	4.86%	4.64%	5.64%	5.03%	0.92%	0.91%	0.64%	1.25%	1.17%
$\rho = 0.3$	Persistent	5.00%	5.05%	4.86%	5.86%	5.28%	1.01%	0.97%	0.62%	1.39%	1.29%
$\rho = 0.3$	Volatile	4.94%	4.96%	4.56%	5.46%	5.07%	1.27%	1.05%	0.92%	1.52%	1.50%
$\rho = -0.3$	Constant	4.93%	4.91%	2.56%**	2.57%**	4.65%	1.03%	1.09%	0.24%**	0.27%**	0.68%
$\rho = -0.3$	Persistent	5.12%	5.14%	2.61%**	2.60%**	4.80%	0.91%	0.88%	0.21%**	0.25%**	0.59%*
$\rho = -0.3$	Volatile	4.93%	5.09%	3.11%**	3.16%**	4.80%	1.54%	1.16%	0.64%	0.67%	1.29%

Rejection of LR_{UC} test statistics: **/*=99%/95% significance.

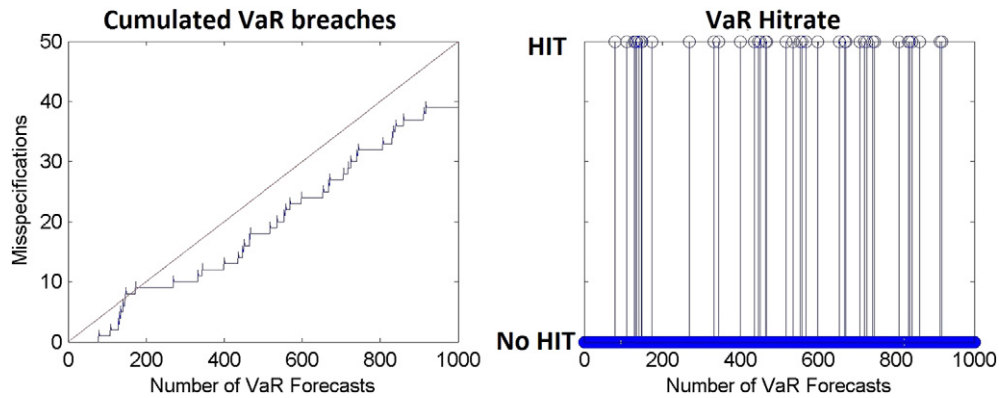

Fig. 3. Empirical 95% Value-at-Risk breaches (1000 obs.). *Left:* Cumulated sum of VaR breaches (Hits). *Right:* Hits over time.

characterized by different patterns of tail dependence. The results are given in the next subsection.

3.2. Tail dependence

The results of analysis regarding tail dependence are given in Table 3 . If the simulated samples exhibit tail dependence, which is modeled via Clayton copula, the application of Gauss, Gumbel and Frank copula leads to inadequate VaR forecasts for all three scenarios ($\theta = 0.5; \theta = 1.5; \theta = 2.5$). As the applied approaches are not able to capture negative tail dependence appropriately, all approaches underestimate the risk in the tail of the loss distribution and lead to empirical failure rates that get rejected by the applied statistical backtesting criterion. Furthermore, the t copula captures appreciably negative tail dependence ($\theta = 2.5$) adequately and results in empirical failure rates that get accepted by the unconditional coverage criterion (LR_{UC}).

If returns exhibit positive tail dependence modeled via Gumbel copula, all copulas lead to adequate empirical failure rates for both 95% and 99% VaR forecasts. As well, if the samples are modeled via Frank copula, the choice of copula does not affect the backtesting results of the 95% VaR forecasts. In contradiction to that, for the 99% VaR forecasts, Gauss, t and Clayton copulas tend to overestimate the risk and lead to overly conservative 99% VaR forecasts, and thus get rejected by the applied backtesting criterion. However, for stronger tail dependence ($\theta = 30$) all copulas get accepted by the backtesting criterion. Interestingly, the choice of copula becomes relevant for VaR forecasts if the sample is described by weak tail dependence modeled by Clayton and Frank copulas. The only copula that leads to appropriate 95% VaR forecasts for all samples is given by the Clayton copula.

3.3. Time varying dependence

Next, to take account of the properties of financial time series, we generate normally distributed return series that are characterized by time varying conditional correlations to study the effect of time varying dependence schemes on the applied copula approaches.

Therefore, we amend the first step of our simulation setup (see Table 1) and model time varying dependence via dynamic conditional correlations. To cover a wide range of joint distributions, we analyze three unconditional correlation settings (strongly positive ($\rho = 0.8$), weakly positive ($\rho = 0.3$), and weakly negative ($\rho = -0.3$)) and for each correlation setting three volatility patterns of ρ (constant, persistent and volatile). The underlying data generating process is a DCC model as introduced by Engle (2002) (see appendix) and the time varying dependence schemes are illustrated in Fig. 4 .

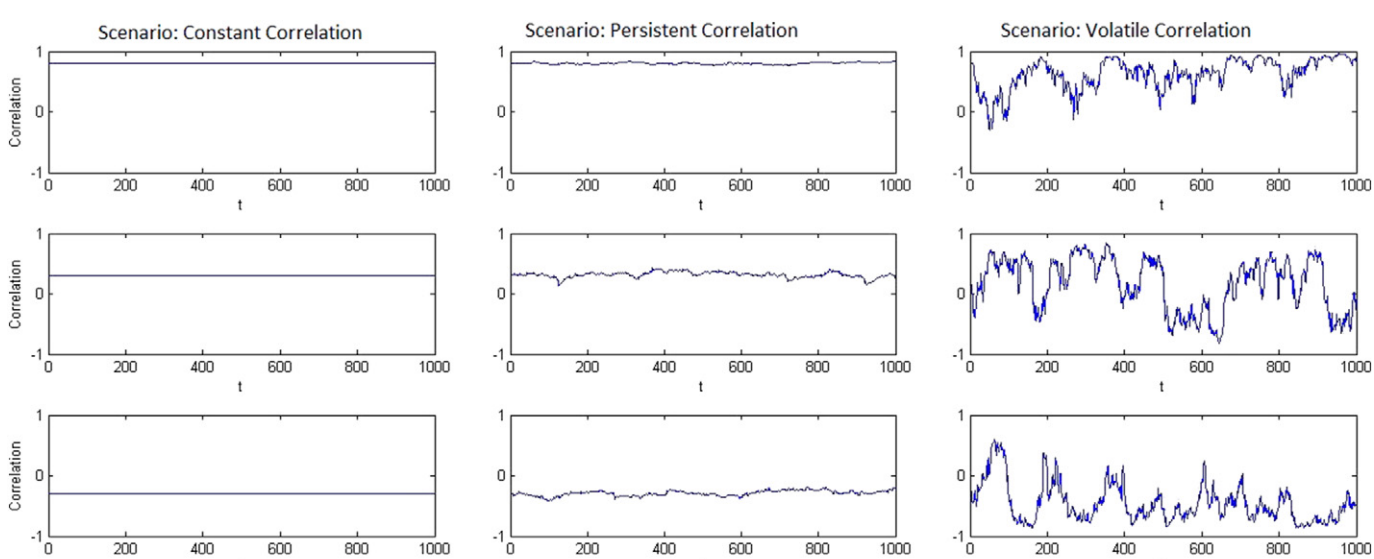

Fig. 4. Correlation scenarios

Our analysis is based on the following parameter values: constant correlation with $\alpha = 0.00$ and $\beta = 1.00$, persistent conditional correlation with $\alpha = 0.01$ and $\beta = 0.97$ and volatile conditional correlation with $\alpha = 0.11$ and $\beta = 0.87$. According to Engle (2009), the medium conditional correlation scenario represents the average parameterization for daily stock returns. Moreover, the parameterization of volatile conditional correlation, which could be observed after Lehman filed bankruptcy, allows us to assess volatile dependence schemes.

Again, for each scenario 10.000 return series covering 1001 observations are generated and the competing copula models are evaluated by the VaR forecasting performance regarding the 1001*th* observation. The results are presented in Table 3.

If the sample is characterized by high correlation ($\rho = 0.8$) only the Frank copula leads to inadequate 99% VaR forecasts for constant, persistent and volatile conditional correlations. It is noteworthy that Clayton copula tends to overestimate the risk and Gumbel copula underestimates the risk at the tail. However, the elliptical copulas are characterized by precise failure rates. For a weak sample correlation ($\rho = 0.3$), no copula gets rejected but again, Clayton copula tends to overestimate the risk and Gumbel copula leads to a higher empirical failure rate regarding VaR forecasts.

If the simulated sample is characterized by a constant sample correlation ($\rho = -0.3$), Clayton and Gumbel fail to adequately capture the downside risk of the simulated assets. Both approaches overestimate the risk and lead to a failure rate that gets rejected by the applied backtesting. As well, both approaches overestimate the risk for samples that are described by persistent and volatile dependence, since 95% VaR forecasts lead to an insufficiently low empirical failure rate. The application of Frank copula to the persistent dependence regime leads to inadequate 99% VaR forecasts. Further, elliptical copulas do lead to acceptable VaR forecasts for all scenarios. It is noteworthy, that both t and Gauss copulas do not account for time variance but lead to adequate results when dependence is time varying. Neither Clayton nor Gumbel and Frank copulas led to acceptable results for all scenarios. Especially, when returns are characterized by negative correlation, Clayton and Gumbel copulas are not able to adequately capture the diversification effect and lead to too conservative VaR forecasts.

4. Conclusion

In this paper, we illustrate the isolated impact of competing copula approaches on daily VaR forecasts. That is, we assess both compatibility and robustness of the applied approaches in the absence of misspecified marginal return distributions.

The conducted simulation analysis reveals that misspecfied dependence schemes impact quality of portfolio VaR and lead to an amount of VaR breaches that deviates from its expected rate by up to 2.5% for 95% forecasts and 1% for 99% VaR forecasts.

Moreover, there is no copula approach that leads to robust 99% VaR forecasts when the data generating process is described by asymmetric tail dependence that is modeled by Archimedean copula approaches.

Due to the fact, that t copulas lead to solid 95% VaR forecasts when the sample is characterized by different degrees of tail dependence as well when dependence is time varying, our results indicate that t copulas describe a sensible choice for applied risk measurement.

However, as this analysis tends to assess the robustness of basic copula concepts with focus on daily VaR forecasts that get evaluated by regulatory standards, we leave the analysis of more elaborated copulas and its theoretical implications to further research. Moreover, as the application of copula approaches to decomposed return series presents a growing field of research (see Reboredo and Riveira-Castro, 2014, Berger, 2015 and the references therein), further research on simulated return scenarios should also take decomposed return series into account.

We implement the assigned dynamic correlation structure following iterative multi-period process.

Let the initial step, for $t = 1$ be presented by:

$$Q_1 = (1 - \alpha - \beta) * \bar{R}, \tag{12}$$

$$R_1 = diag\left\{Q_1^{1/2}\right\} Q_1 diag\left\{Q_1^{1/2}\right\}, \tag{13}$$

$$(\epsilon_{(i,1)}^{sim} \quad \epsilon_{(j,1)}^{sim}) = (\epsilon_{i,1}\epsilon_{j,1}) * \sqrt{R_1} \tag{14}$$

whereas $\bar{R}$ represents the sample correlation and α and β the DCC parameters. Based on this step, we proceed iteratively by modeling the dependence structure for the following period. So to say, conditionally on period t, we model the dependence structure for $t + 1$:

$$Q_{t+1} = (1 - \alpha - \beta) * \bar{R} + \alpha * (\epsilon_t^{sim\text{‘}}\epsilon_t^{sim}) + \beta(Q_t) \tag{15}$$

$$R_{t+1} = diag\left\{Q_{t+1}^{1/2}\right\} Q_{t+1} diag\left\{Q_{t+1}^{1/2}\right\}, \tag{16}$$

$$(\epsilon_{t+1,1}^{sim} \quad \epsilon_{t+1,1}^{sim}) = (\epsilon_{t+1,1} \quad \epsilon_{t+1,1}) * \sqrt{R_{t+1}}. \tag{17}$$

In this setup β represents the persistency of the process.

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